EXPERIMENT: 30

CREATING THE APPLICATIONS USING TCP CHAT CLIENT AND CHAT SERVER IN JAVA/C

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <sys/wait.h>

int main()
{
    int server, client, new_socket;
    int port = 8080;
    char message[100];
    struct sockaddr_in server_addr, client_addr;
    socklen_t addr_size = sizeof(client_addr);

    printf("TCP CHAT APPLICATION\n");
    printf("--------------------\n");

    server = socket(AF_INET, SOCK_STREAM, 0);

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(port);
```

```
bind(server, (struct sockaddr *)&server_addr, sizeof(server_addr));
listen(server, 1);

printf("Server started...\n");
printf("Waiting for client...\n");

client = fork();

if (client == 0)
{
    /* CLIENT */

    sleep(1);

    int sock = socket(AF_INET, SOCK_STREAM, 0);

    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    connect(sock, (struct sockaddr *)&server_addr,
            sizeof(server_addr));

    printf("\nCLIENT CONNECTED\n");

    strcpy(message, "Hello Server");
    send(sock, message, strlen(message) + 1, 0);

    printf("Client: %s\n", message);
```

```
    recv(sock, message, sizeof(message), 0);

    printf("Server: %s\n", message);

    close(sock);
}
else
{
    /* SERVER */

    new_socket = accept(server,
                        (struct sockaddr *)&client_addr,
                        &addr_size);

    printf("\nSERVER: Client connected!\n");

    recv(new_socket, message, sizeof(message), 0);

    printf("Server received: %s\n", message);

    strcpy(message, "Hello Client");

    send(new_socket, message, strlen(message) + 1, 0);

    printf("Server: %s\n", message);

    close(new_socket);

    wait(NULL);
```

```
    }

    close(server);

    return 0;
}
```